Organic Cume, December 4, 2015

Below are reaction names or descriptions of reactions that are encountered in a typical undergraduate organic chemistry class. Give a specific example for each when the reactant(s) is not specified (no generalizations with "R" groups) and then show a detailed mechanism using your reactant(s) and product(s), including electron pushing, for each.

1. SN1 reaction with alcohols

2. Butadiene reaction with hydrogen halides (kinetic vs thermodynamic control)

3. Simmons-Smith reaction

4. Peroxyacid reaction (Z)-2-butene

5. Hydroboration/Oxidation alkenes

6. Trans alkenes from alkynes

7. Bromination of benzene

8. Williamson Synthesis

9. Acidic conditions for the hydration of aldehydes and ketones

10. Cyanohydrin formation